

Corrosion + Rust Removal

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Find the battery nobody wanted, then run it backwards

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 5 — Corrosion and Electrolytic Rust Removal

Hook: Rust is a battery you did not want. Today we measure the voltage that drives it, make iron rust faster on purpose, protect it, and open the cells that spent the night running the reaction backwards.

Learning outcomes

- **Predict** the voltage of a metal pair from the electrochemical series, then **measure** it and account for the gap
- Measure the **short-circuit current** of a galvanic couple and convert it, via Faraday's law, into a **corrosion rate in grams per year**
- Explain galvanic corrosion and sacrificial protection, and identify which metal in a pair is at risk
- Interpret the overnight **electrolytic rust removal** cells started in Session 4
- Place all five sessions on a single axis: **spontaneous chemistry making electricity**, or **electricity forcing chemistry**

Session plan (90 min)

Time	Activity	Reference
0–15 min	Part A — Reveal the derusted nails and the corrosion jars	experiment.md — Part A
15–30 min	Activity series, galvanic couples, sacrificial protection	lecture.md §3
30–55 min	Part B — Metal-pair cells: predict, measure, then measure the current	experiment.md — Part B
55–68 min	Part C — Corrosion jars A / B / C	experiment.md — Part C
68–78 min	Part D — The week on one table	experiment.md — Part D
78–90 min	Capstone challenge — applied problems, then the closing	lecture.md §5

Key reactions

Rusting, as two half-reactions happening on the same piece of iron:

Anodic sites: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$

Cathodic sites: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$

Fe^{2+} is then further oxidized and precipitates as hydrated iron oxides — rust.

Galvanic couple (Fe touching Cu in saltwater):

Iron becomes the anode of a real cell and corrodes faster.

Sacrificial protection (Fe touching Zn):

$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ instead. Iron is now the cathode and is protected.

Electrolytic derusting:

A power supply forces the iron to be the cathode. Oxide is reduced and lifted; H_2 evolves.

Multi-day setup this session depends on

Demo	Must be started	Revealed
Corrosion jars A / B / C	Session 1 — see shared/multi-day-setup.md	Session 5
Electrolytic rust removal	Session 4 , end of class	Session 5, opening

If the jars were not started on Day 1, read the fallback in preparation.md before class rather than during it.

Status

- ☐ Derusting cells confirmed still powered this morning
- ☐ "Before" photos located and ready to display
- ☐ Corrosion jars A / B / C available (or fallback plan chosen)
- ☐ Metal strips cleaned: Zn, Fe, Cu, Al, and Mg if available
- ☐ 5–10% saltwater mixed and labeled "**corrosion demos only**"
- ☐ Capstone worksheets printed

Session 5 — Preparation

Check this first, this morning

- [] **Are the derusting cells still powered?** Touch the supply, look for bubbling, confirm the nail is dark rather than orange. If the power died in the night, read the fallback below before class rather than during it.
 - [] **Is the control nail there, unpowered, in the same solution?** Without it there is no experiment.
 - [] **Can you find the before photos?** Have them open and ready to display.
-

Session 4 handoff (the critical dependency)

These cells must be running before students leave Session 4:

Item	Detail
Electrolyte	Washing soda preferred; baking soda acceptable — never NaCl
Cathode (-)	Rusty iron
Anode (+)	Sacrificial steel or graphite (not stainless)
Control	Equally rusty nail, same solution, no power
Power	Mains supply or 5 V USB brick. A 9 V alkaline is flat in about two hours
Ventilation	Jars open, room ventilated — these cells make hydrogen all night

If the reveal failed

It happens, usually because a battery died or a clip came loose. Options, in order of preference:

1. **Run a fast cell live.** Wire a fresh rusty nail at the highest safe current at the start of class and come back to it at the 60-minute mark. An hour at 0.5 A visibly changes a lightly rusted nail.
2. **Have a prepared example.** Keep one successfully derusted nail from your pilot run, sealed in a bag. Show it, and be straightforward that it is from the pilot.
3. **Teach the failure.** Ask the class to diagnose it: dead battery, loose clip, reversed polarity, too little electrolyte. Then check the anode — if it is heavily eaten, current definitely flowed, and the question becomes where the effect went. This is a genuinely good fifteen minutes and students respect it more than a smooth demo.

Whatever happens, write down what went wrong in `shared/teaching-log.md` while it is fresh.

Before class (instructor)

- [] Mix **5–10% saltwater** and label it "**CORROSION DEMOS ONLY — NOT FOR ELECTROLYSIS OR DERUSTING**"
- [] Physically separate the saltwater from anything left over from Session 4
- [] Sand all metal strips to bare metal, especially the **aluminium**, and keep them dry until use
- [] Confirm every multimeter has a **working mA range** — Part B3 needs it, and a blown fuse reads zero

- [] Pilot the metal pairs yourself and record what your saltwater actually gives; the numbers in instructor.md are typical, not guaranteed
- [] Check the corrosion jars: are the metal contacts still firm? A slipped wire makes jar B look like jar A
- [] Print E° tables with the **predictions column blank**
- [] Print capstone worksheets and week-comparison tables
- [] Locate photos of a ship's hull anode and a water-heater rod

Multi-day setup this session depends on

The corrosion jars need **four to five days** to develop. They belong at the end of Session 1, not here — see shared/multi-day-setup.md.

Jar	Setup
A	Iron nail alone in saltwater
B	Iron nail in firm contact with copper
C	Iron nail in firm contact with zinc, or a galvanized nail

Fill each jar **half full** so the metals cross the waterline. Keep them where students can look at them daily — the running commentary through the week is half the value.

If the jars were not started: set them up today anyway as a demonstration of method, and use the photograph fallback for the results. Then put a note in the teaching log so next year's Day 1 includes them.

Day-of timeline

When	Task
Before students	Verify cells still powered; disconnect nothing yet
0 min	Part A — reveal, with control and before-photo
15 min	Activity series, galvanic couples, sacrificial protection
30 min	Part B — predictions collected before meters are handed out
55 min	Part C — jars A / B / C
68 min	Part D — week comparison table
78 min	Capstone worksheet, then the closing
90 min	Collect worksheets; photograph everything for next year

Troubleshooting

Issue	Cause	Fix
Treated nail looks like the control	Power died, clip loose, or too little time	See the fallback options above
Nail is <i>worse</i> than when it started	Polarity reversed — it was the anode	Excellent teaching moment; show the class the wiring
Bath smells of chlorine	Salt was used	Stop, ventilate, dispose, remake with washing soda
Metal-pair voltage near zero	Metals touching in solution, or an oxide layer	Separate them; sand both
Aluminium reads lower than iron	Expected — passive oxide film	This is the lesson, not a fault
Voltages well below prediction	Non-standard conditions; oxygen reduction at the cathode	Expected. See instructor.md
Short-circuit current reads zero	Meter not on mA, or a blown fuse	Check the fuse
Jar B looks identical to jar A	Contact slipped — no electron path	Re-clamp; note it for next year
Jars barely rusted at all	Started too late, or filled too full	Use the fallback photo; start jars on Day 1 next year

Open questions / notes for next year

- Overnight power source used, and hours run:
- Was the reveal convincing?
- Measured voltages: Zn-Cu _ / **Fe-Cu** / Al-Cu _ / **Mg-Cu**
- Zn-Cu short-circuit current:
- Date the corrosion jars were started, and whether that was early enough:
- Which capstone questions worked, and which fell flat:
- Did the session finish on time, or did the reveal overrun?

Session 5 — Materials

Part A — The reveal (cells started in Session 4)

- ☐ The overnight derusting cells, **still powered this morning** — check before students arrive
- ☐ The **unpowered control nail**, in the same solution, clearly labeled
- ☐ Printed or on-screen "**before**" photos
- ☐ Forceps or gloves
- ☐ Access to running water for rinsing
- ☐ Stiff brush or wire brush, and paper towels
- ☐ Tray to work over — the sludge stains

Part B — Metal-pair cells

Per group

- ☐ Small jar, ~100 mL
- ☐ **Saltwater 5–10% w/v** (25–50 g NaCl per 500 mL), labeled "**corrosion demos only**"
- ☐ Copper strip or heavy wire
- ☐ Iron nail or steel strip
- ☐ Zinc strip or galvanized nail
- ☐ Aluminium strip or heavy foil
- ☐ Multimeter with **both** DC volts and DC mA
- ☐ Alligator clips
- ☐ Sandpaper or steel wool
- ☐ Paper towels
- ☐ Calculator
- ☐ **Printed E° table with the predictions column blank** — experiment.md Part B1

Optional but worth having

- ☐ Magnesium ribbon — the biggest voltage of the set and a good contrast with aluminium
- ☐ A second nail already sanded to bare metal, for comparison

Part C — Corrosion jars

- ☐ Jars A, B and C, **started in Session 1** — see shared/multi-day-setup.md
- ☐ A photograph of well-developed jars, as a fallback if this year's underperform

Part D and capstone

- ☐ Printed week-comparison tables
- ☐ Printed capstone worksheets — lecture.md §5

- [] Photos of sacrificial anodes: a ship's hull, a water-heater rod

Instructor display items

- [] A galvanized nail and a plain iron nail, side by side, to pass around
- [] Activity series written large, to stay on the board all session

Explicitly excluded

- [] ~~NaCl in the derusting bath~~ — washing soda or baking soda only
- [] ~~Stainless steel anodes~~ — can release chromium compounds into the bath

Pilot notes — fill these in for next year

Parameter	Value
Overnight power source actually used	
Hours the derusting cells ran	
Was the difference from the control obvious?	
Measured Zn-Cu voltage	_____ V
Measured Fe-Cu voltage	_____ V
Measured Al-Cu voltage	_____ V
Zn-Cu short-circuit current	_____ mA
Date the corrosion jars were started	
Were jars B and C visibly different from A?	

Session 5 — Experiment: Corrosion and Electrolytic Rust Removal

Instructors: reveal protocol, why measured voltages fall short of predictions, jar narratives — instructor.md.

Session arc

1. **Part A** — Open the overnight derusting cells (the reveal)
2. **Part B** — Metal-pair cells: **predict** the voltage, measure it, then measure the **current** and convert it into a corrosion rate
3. **Part C** — Corrosion jars A / B / C
4. **Part D** — The whole week on one table

Solutions used today

Saltwater — for corrosion and galvanic demos only

Ingredient	Amount
Table salt (NaCl)	25–50 g per 500 mL
Water	500 mL tap water

That is roughly **5–10% w/v**. Label it: "**CORROSION DEMOS ONLY — NOT FOR ELECTROLYSIS OR DERUSTING.**"

Today is the one day of the week salt is allowed in the room, and it is allowed only because nothing here is being driven hard enough to make chlorine. Keep it physically apart from the derusting jars.

Derusting bath (mixed in Session 4)

Ingredient	Amount
Washing soda (Na_2CO_3), preferred	1 tbsp (~15 g) per 250 mL warm water
or baking soda (NaHCO_3)	1 tbsp per 250 mL warm water

Part A — The reveal (0–15 min)

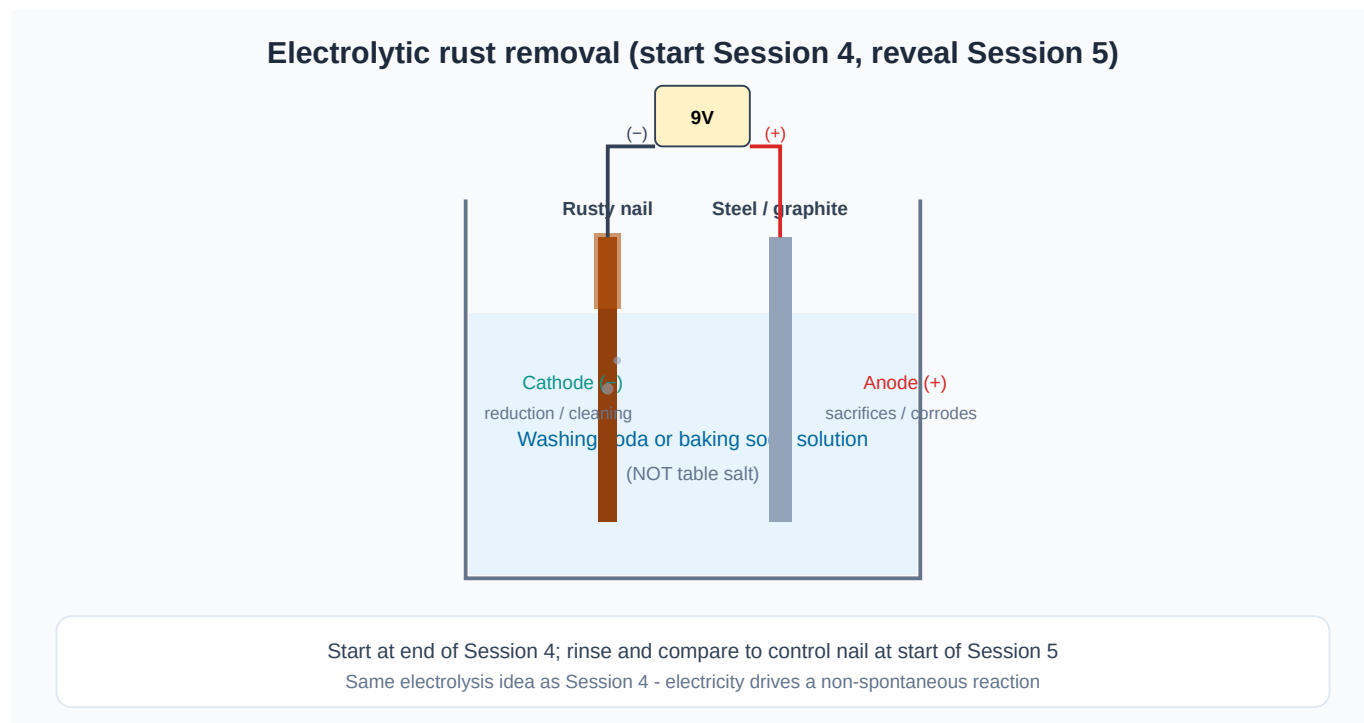


Figure 1 — Cells started at the end of Session 4: rusty iron on the cathode (–), sacrificial anode on (+), in washing soda solution.

Do this before any teaching. It is the best visual of the week and it buys you fifteen minutes of attention.

Procedure

1. **Disconnect the power before touching anything.** Every time, without exception.
2. Lift the treated nail out with forceps or gloved fingers and rinse it under running water.
3. A stiff brush or a wipe with a paper towel removes the loosened black film. Say what you are doing and why — the electrochemistry loosens the oxide, it does not vaporise it.
4. Put the treated nail, the **unpowered control nail**, and the **before photo** side by side.
5. Look at the rest of the cell too, not just the nail: the anode is pitted and dark, the solution is cloudy orange-brown, and there is sludge at the bottom. That material is the rust, relocated.

Observation sheet

Sample	Electrolyte	Powered?	Appearance after rinse	Cleaner than control?
Treated nail	Washing soda	Yes		
Control nail	Same solution	No		
Sacrificial anode	—	Yes		
The bath itself	—	—		

The control is the point. Soaking a rusty nail in alkaline solution overnight does something all by itself, and without the control you cannot claim the electricity did anything.

Discussion prompts

- Which electrode was the cathode? What happens at a cathode?
- How is this the same as Session 4's electrolysis? (*Same driven cell, same hydrogen at the negative electrode.*)
- Where did the rust actually go?
- What would have happened if we had reversed the clips? (*We would have dissolved the nail. This is the entire experiment in one wire swap.*)

Part B — Metal-pair cells: predict, then measure (30–55 min)

Session 5 - Galvanic cell (Zn + Cu in saltwater)

$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (oxidation at anode) | Spontaneous - produces electricity like Session 1
More active metal becomes anode; less active metal becomes cathode

Figure 2 — Two dissimilar metals in saltwater make a cell. Zinc corrodes as the anode; copper is the cathode and is protected.

This is Session 1's fruit battery with the fruit removed and the chemistry made explicit — and this time we predict the answer before we measure it.

B1 — Predict first (5 min, before touching any equipment)

Standard reduction potentials, in volts, versus the standard hydrogen electrode:

Half-reaction	E° (V)
$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$	-2.37
$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	-1.66
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.76
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	-0.44
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.00

Half-reaction	E° (V)
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	+0.34
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$	+0.80

The rule: the metal with the **more negative** value is the one that corrodes. It is the anode.

$$\text{Predicted cell voltage} = E^\circ(\text{cathode metal}) - E^\circ(\text{anode metal})$$

Fill in the predictions column **before** you get a multimeter.

B2 — Measure the voltage

1. Pour ~50–100 mL of **5–10% saltwater** into a small jar.
2. Sand both metal strips lightly if they are dull or oxidized. Aluminium especially — it carries an invisible oxide film that will change your answer.
3. Immerse both metals to a similar depth, **not touching each other** in the solution.
4. Multimeter on DC volts. Touch one probe to each metal.
5. If the reading is negative, **swap the probes and note it** — the metal now on the red probe is the cathode, and the other one is corroding.
6. Wipe the electrodes between trials.

B3 — Measure the current (this is the new part)

Open-circuit voltage tells you *whether* corrosion is favoured. It says nothing about *how fast*. For that you need current.

1. Switch the meter to **DC mA** and connect it **directly between the two metals**, so it completes the circuit.
2. The reading falls quickly at first as the surfaces polarize. Take the value after about 30 seconds and note that it is drifting.

That current is the corrosion rate, in electrons per second. It is the same Faraday's law from Sessions 3 and 4, pointed at a reaction nobody wanted.

Data table

Metal 1	Metal 2	Predicted V	Measured V	Which metal is (+)?	Anode (corrodes)	Short-circuit I (mA)
Zn	Cu	+1.10				
Fe	Cu	+0.78				
Al	Cu	+2.00				
Zn	Fe	+0.32				
Mg	Cu	+2.71				

B4 — Turn the current into a corrosion rate

For zinc, which needs 2 electrons per atom ($M = 65.4 \text{ g/mol}$):

$$\text{mass lost per second} = I \times 65.4 / (2 \times 96,485)$$

Useful benchmark, worth putting on the board:

A steady 1 mA eats about 11 grams of zinc per year.

Quantity	Your value
Short-circuit current, Zn-Cu	_____ mA
Zinc consumed per hour	_____ g
Zinc consumed per year at this rate	_____ g
A 500 g sacrificial anode would last	_____ years

That last line is real engineering. Sizing the zinc block on a ship's hull is exactly this calculation, run with a current measured on the actual structure.

What you will find, and why it is interesting

Your measured voltages will be lower than your predictions — sometimes much lower. This is not sloppy work. Three reasons, and each is worth naming:

Reason	What it does
The E° table assumes 1 M metal-ion solutions at 25 °C. Our saltwater contains essentially no Zn ²⁺ or Cu ²⁺ to start with	Shifts every potential
The reaction actually happening at the copper is oxygen reduction , not Cu ²⁺ → Cu. The table's copper line is the wrong half-reaction for this cell	The largest single reason
Aluminium carries a tough passive oxide film that stops it behaving like bare aluminium at all	Al almost always underperforms badly — often below iron

Aluminium is the star of this section. It is predicted to be violently reactive, second only to magnesium, and it will probably read lower than iron. That gap is the whole reason aircraft and window frames are made of it. Ask the class to explain it before you do.

Part C — Corrosion jars (55–68 min)

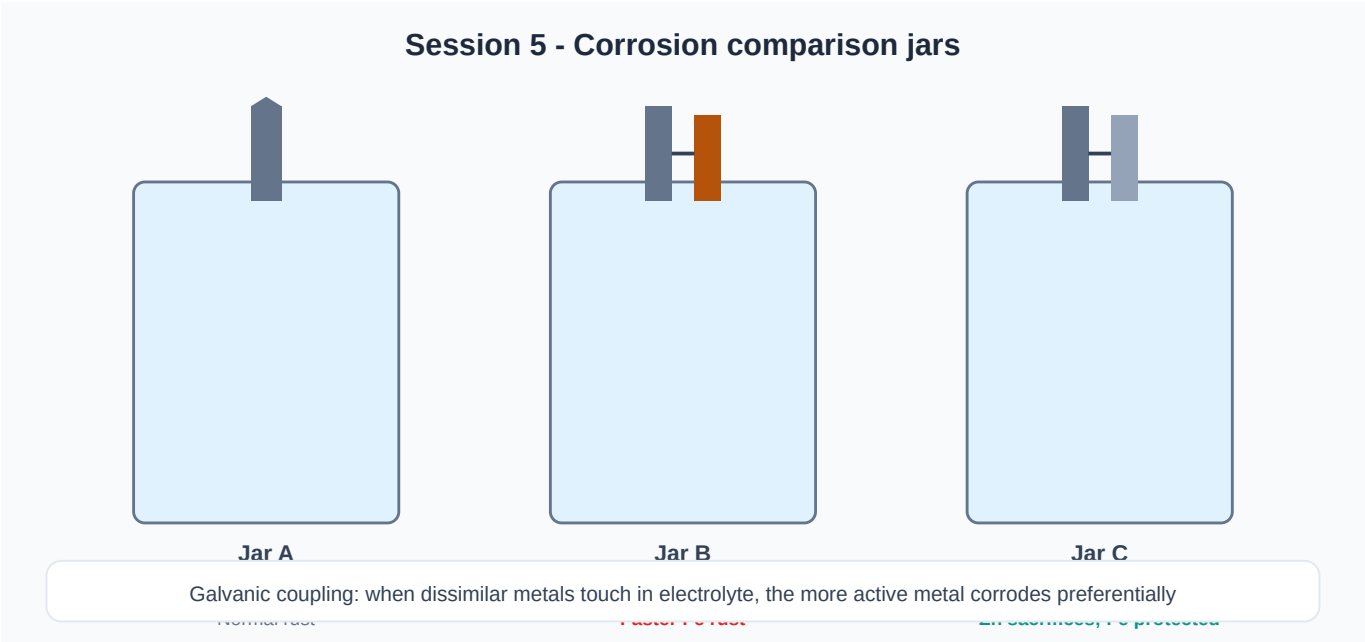


Figure 3 — Jar B: iron rusts faster when coupled to copper. Jar C: zinc protects the iron by corroding instead.

The jars (started Session 1 — see [shared/multi-day-setup.md](#))

Same 5–10% saltwater in each, about 100–150 mL, and each jar only **half full** so the metals sit at the waterline where oxygen is plentiful.

Jar	Contents	Expected after four days
A	Iron nail alone	Moderate, even rust
B	Iron in firm contact with copper	Noticeably worse rust on the iron
C	Iron in firm contact with zinc (or a galvanized nail)	Iron largely clean; the zinc dulls and powders

Contact is not optional. The metals must physically touch or be clipped together, or there is no electron path and no couple. A jar where the wire has slipped shows nothing, and it is the most common way this demo fails.

Today's task

1. Line up A, B and C. Look at the **iron** in each, not the jar as a whole.
2. Note where the rust concentrates: at the waterline, and near the contact point.
3. Rank the three jars by how badly the iron has corroded.
4. Explain the ranking using your Part B data — you have just measured the voltages that drive exactly these couples.

Observation sheet

Jar	Iron appearance	Partner metal appearance	Which metal corroded?	Matching pair from Part B
A		—		—

Jar	Iron appearance	Partner metal appearance	Which metal corroded?	Matching pair from Part B
B		Cu		Fe-Cu, _____ V
C		Zn		Zn-Fe, _____ V

The question to end on

Jar A had no partner metal at all. So why did it rust?

Because a single piece of iron is **already** a collection of tiny cells: impurities, stressed regions, and scratches are slightly anodic, and areas with better oxygen access are cathodic. A bent nail rusts fastest at the bend for exactly this reason. You do not need two metals to corrode — two metals just make it worse and make it predictable.

Part D — The whole week on one table (68–78 min)

Students fill this in from memory, in groups, then compare.

	Fruit battery	Copper plating	Silver plating	Water electrolysis	Rust in jar B	Derusting
Session	1	2	3	4	5	5
Needs external power?						
Produces electricity?						
Spontaneous or driven?						
What is oxidized?						
What is reduced?						
Metal moved?						
Gas produced?						
Useful or unwanted?						

Answer key (instructor)

	Fruit battery	Copper plating	Silver plating	Electrolysis	Rust (jar B)	Derusting
Needs external power?	No	Yes	Yes	Yes	No	Yes
Produces electricity?	Yes	No	No	No	Yes*	No
Spontaneous or driven?	Spontaneous	Driven	Driven	Driven	Spontaneous	Driven
Oxidized	Zn	Cu anode	Water at the graphite anode	Water → O ₂	Fe	Anode metal

	Fruit battery	Copper plating	Silver plating	Electrolysis	Rust (jar B)	Derusting
Reduced	H^+ / O_2 at Cu	$\text{Cu}^{2+} \rightarrow \text{Cu}$	$\text{Ag}^+ \rightarrow \text{Ag}$	Water $\rightarrow$ H_2	$\text{O}_2 \rightarrow \text{OH}^-$	Iron oxide, and $\text{H}^+ \rightarrow \text{H}_2$
Metal moved?	No	Yes	Yes	No	Yes, into solution	Yes, off the surface
Gas produced?	Trace	Little	Little	Yes, both	No	Yes, H_2
Useful or unwanted?	Useful	Useful	Useful	Useful	Unwanted	Useful

* Jar B genuinely generates electricity — that is precisely the problem. Corrosion is a battery discharging itself for free, using your bridge as the anode.

The single sentence the week has been building toward: every one of these is the same process. Electrons leave something and arrive somewhere else. The only questions are which direction, and whether we are paying for it or paying to stop it.

Safety

- Saltwater corrosion demos: wash hands afterwards; gloves optional
 - Keep saltwater **physically apart** from the derusting jars and from anything left over from Session 4
 - Derusting: washing soda or baking soda only — **never NaCl**
 - Disconnect powered cells before touching electrodes, and whenever the room is unsupervised
 - Rusty hardware is sharp and often tetanus-relevant. Forceps or gloves, and no bare-handed scrubbing
 - No open flames near any gas-evolving setup
-

Experiment status

- [] Derusting cells still powered this morning; before photos located
- [] Control nail present and clearly labeled
- [] Saltwater mixed and labeled for corrosion use only
- [] Corrosion jars available, contacts verified intact
- [] Metal strips sanded and labeled
- [] E° prediction table printed with the predictions column **blank**
- [] Capstone worksheets printed

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

The three rules that matter most

1. **Salt belongs in exactly one place this week.** NaCl is used only for the Session 5 corrosion jars, where nothing is driven hard enough to make chlorine. It is **forbidden** in every electrolysis bath and in the derusting cells, where chloride is oxidized at the anode to **chlorine gas**. This is a competing half-reaction, not an arbitrary rule.
2. **Any cell left running unattended must be open and ventilated.** Electrolysis and derusting cells evolve hydrogen continuously. An overnight derusting cell can generate a couple of litres of it. Open jars, ventilated room, no ignition sources, nothing sealed. Ever.
3. **Disconnect power before touching anything wet.** Every time, as a ritual, especially at the Session 5 reveal.

General rules

1. **No eating experimental materials.** Fruit, potatoes and liquids used in experiments are no longer food, and this includes the ones that started out edible.
2. **Ammeters go in series.** A meter on a current range clipped across a supply is a short circuit and will blow its fuse at best.
3. **Avoid short circuits generally.** Batteries heat up fast when shorted.
4. **Collect metal-salt waste.** Copper and silver solutions are not poured down the drain.
5. **Water choice:** tap water is fine for most baths. **Distilled is required for the Session 3 silver bath**, so chloride does not precipitate silver, and is worth having for the first rung of the Session 4 conductivity ladder.
6. **Goggles for all wet work.** Gloves additionally for silver nitrate.
7. **Rusty hardware is sharp** and often outdoors-dirty. Forceps or gloves; no bare-handed scrubbing.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails and wires	Supervise handling; wash hands after handling metals
2	Copper sulfate solution	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves and goggles; stains skin and clothing near-permanently; small volumes (30–40 mL); collect all waste; no ammonia, ever
4	Gas evolution; hydrogen ignition	No NaCl; ammeter in series; U-tube side arms open; balloon and flame instructor only
5	Corrosion demos; overnight derusting	Washing soda only, never NaCl; disconnect before handling; jars open and ventilated overnight

Silver nitrate and ammonia — Session 3

Never add ammonia, or any ammonia-containing cleaner, to a silver nitrate solution. Ammoniacal silver solutions (Tollens' reagent) deposit **silver nitride and related compounds** on standing, which are friction- and shock-sensitive explosives. The hazard is not the fresh solution — it is any residue left in a bottle, on a stirrer, or around a cap.

This matters because the "silver mirror" recipes students will find online use exactly this chemistry. Say so out loud during the Session 3 safety brief, before someone tries it at home.

It also does not even work for our purpose: ammonia complexes copper about as strongly as it complexes silver, so it does nothing to suppress the displacement reaction we are trying to control.

Hydrogen — Session 4, instructor only

The balloon demo is optional. The chemistry is fully demonstrated by the splint tests without any of this risk, so skip the balloon entirely if the venue forbids open flame or if you are at all unsure.

If you do run it:

- Feed **cathode (H₂) gas only** into the balloon. The anode vents separately to the room.
- **Never a deliberate H₂/O₂ mixture.** Pure hydrogen in air *deflagrates* — it burns at the surface and gives a bark. A stoichiometric mixture **detonates**, which is a different event entirely and not a classroom demonstration.
- **Never a rigid or sealed container.** A balloon tears; glass becomes shrapnel.
- Modest inflation only.
- Tie off, then carry it **well away** from the electrolysis cell, which is still making hydrogen.
- Long lighter at arm's length; area cleared; extinguisher present and its location known.
- Warn the room. The bang is loud, sudden, and some students find it distressing.

Splint tests — the low-risk alternative

Gas	Test	Result
H ₂	Lit splint at the mouth of the tube	Squeaky pop
O ₂	Glowing (not flaming) splint inserted	Ember relights

A few millilitres of gas in a small test tube is plenty. Keep hydrogen tubes **mouth-down** until the moment of testing. TA-supervised; students may hold the tube, an adult holds the flame.

Waste

Waste	Disposal
Copper sulfate solution	Labeled waste container; not down the drain
Silver nitrate solution and rinses	Labeled waste container; not down the drain. Drop a coil of bare copper wire in — it strips the silver back out as metal for recovery
Sodium sulfate electrolyte	Drain is acceptable; dilute well

Waste	Disposal
Washing soda derusting sludge	Let solids settle, bag the sludge, dilute the liquid to drain
Saltwater corrosion jars	Drain; dispose of rusty metal in solid waste